CS 200 FINAL PROJECT



AVA TRAISCI

The dataset we will be looking at is:

ONLINE SHOPPERS PURCHASING INTENTION DATASET



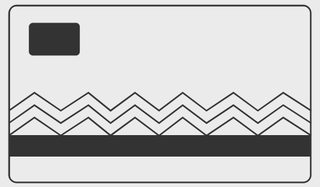






From the UCI Repository

What Is This Data Set?

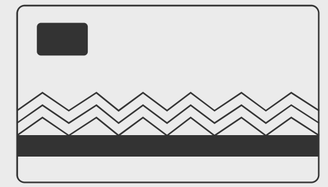


- This dataset holds a bunch of user information that occurs when a user visits a website

- The dataset includes “Revenue”, which is a binary feature
  - A “True” entry for this features means that when a user visited a website, there was a transaction that was made by the user
  - A “False” entry for this features means that when a user visited a website, there was no transaction that was made by the user



Examples of features in this dataset:



- "Administrative" , “Administrative_Duration" , "Informational", "Informational_Duration", "ProductRelated", "ProductRelated_Duration"
  - All of these features explain the types of websites visited by the user, and how long they spent on these pages



- "Bounce Rate", "Exit Rate", "Page Value"
  - These features explain how a user leaves the website

- “SpecialDay”, “Month”, “OperatingSystem”, “Browser”, “Region”, “TrafficType”, “VisitorType”, and “Weekend”
  - These are some more more self explanatory features included in this set

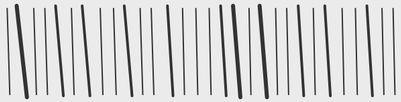


My Goal:





Create a model that can predict if a user’s web visit resulted in them purchasing a product



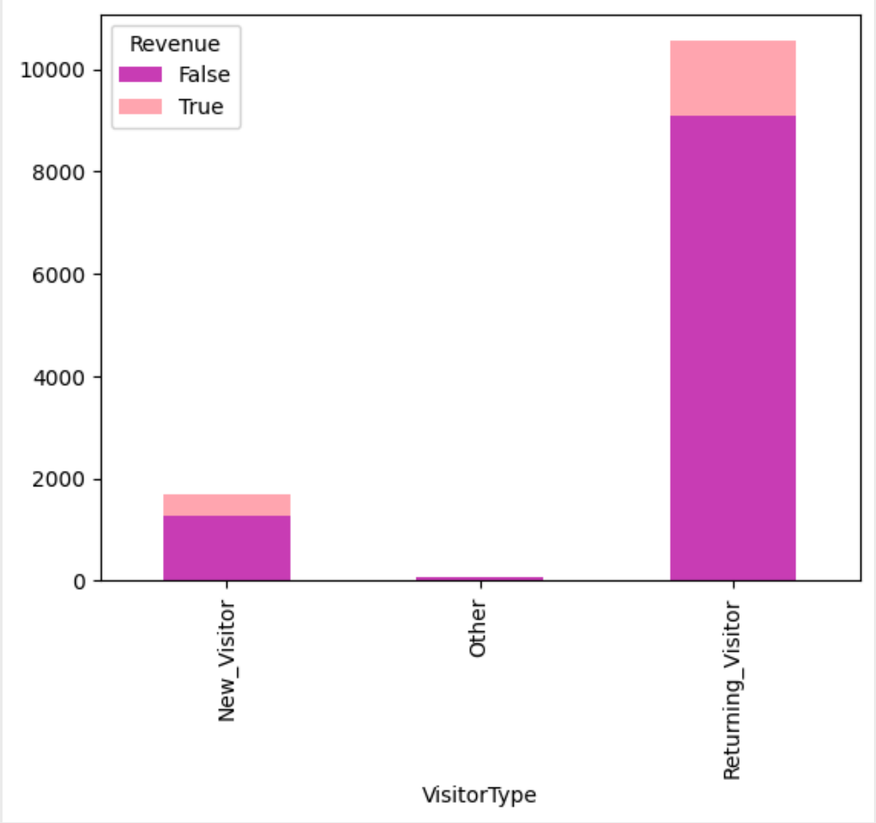




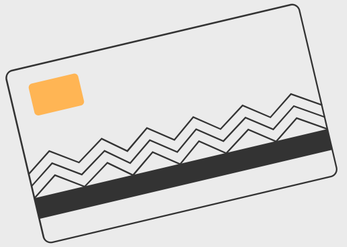
This is important as marketing teams would hopefully be able to use this model to determine what attributes lead to a user purchasing a product

Exploring Data

- Determined there were no missing values in the dataset
- Target was very unbalanced (84.5% of the entries were “False”, 15.5% were “True”)
- Pairplot showed initial associations between "Exit Rates" and "Bounce Rates", and "Product Related Duration" and "Informational"
- Stacked bar chart between “Revenue” and “VisitorType” showed there are more transactions that occur when the user has previously visited the page (the user is a returning visitor) than when a user visits the page for the first time
- Chi-sq and Ttest showed strong association between “Month” & “Revenue” and “ExitRates” and “Revenue”



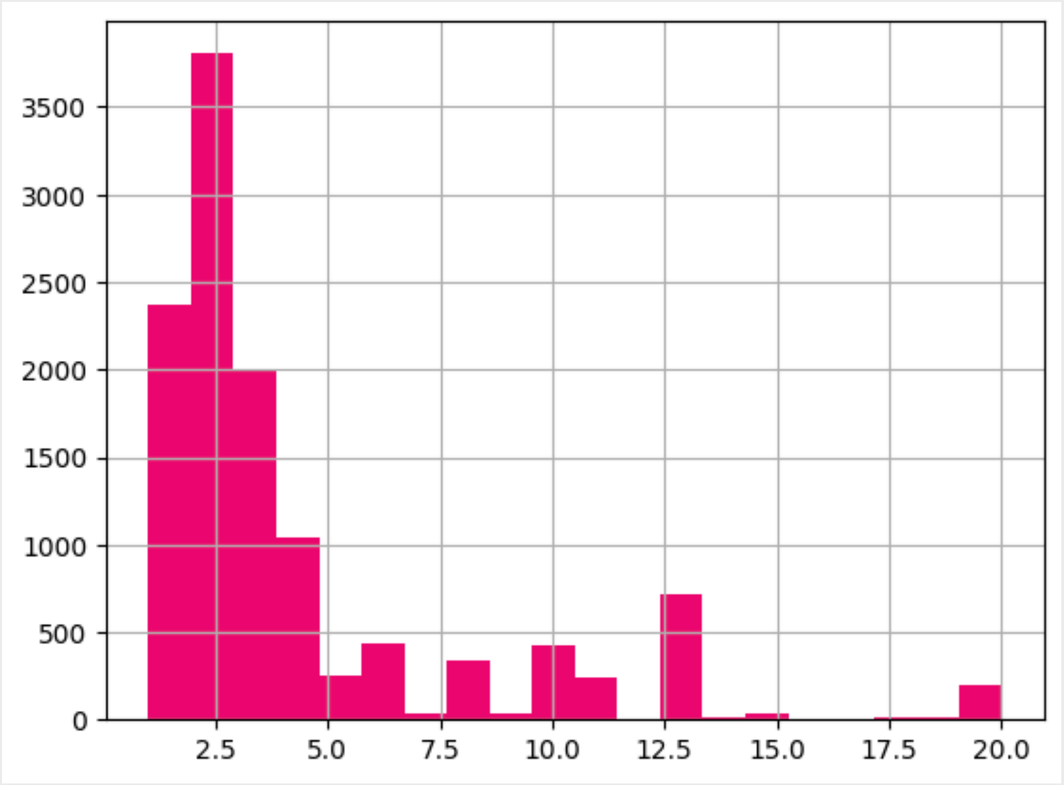
Data Cleaning



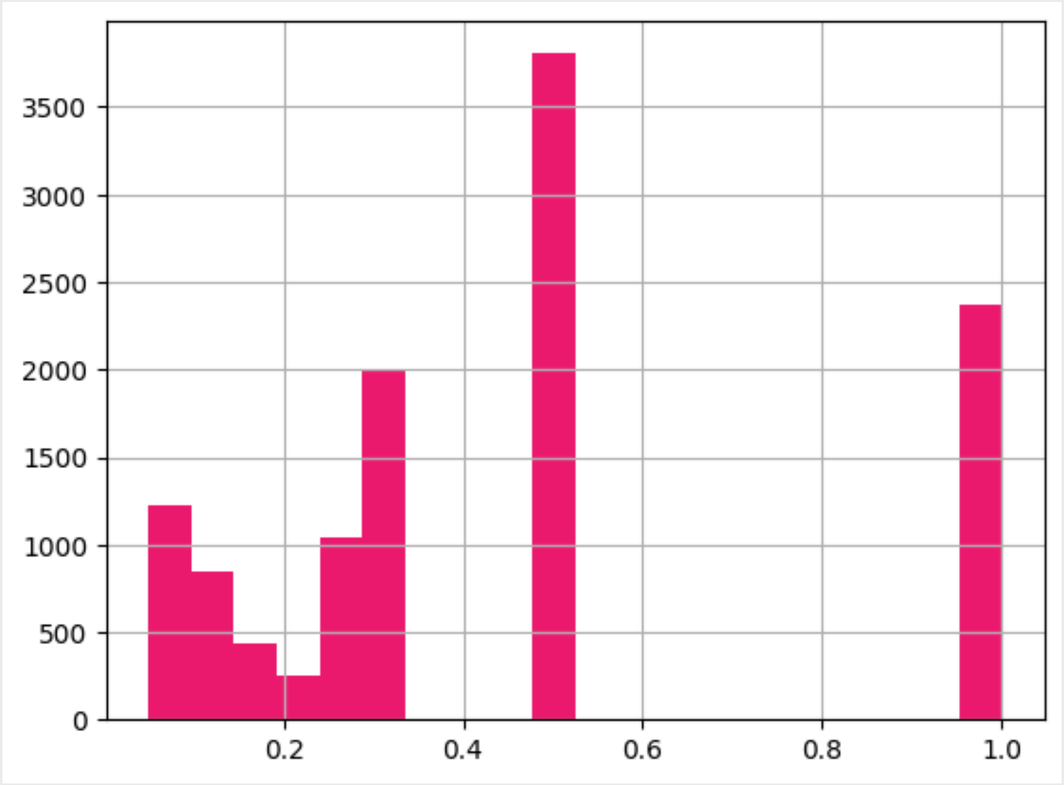


- Fixed unbalanced target issue by undersampling the majority class (“False”) by 19%
- Transformed right skewed features
  - “TrafficType” with 1/x transformation
  - “Browser” with log transformation
- One hot encoded “VisitorType” and “Month” so that these features could be processed
- Feature engineered Visitor Type features to remove redundancy
- Removed "Administrative_ Duration" feature as it was not useful

Data Cleaning



“TrafficType” before 1/x



“TrafficType” after 1/x

Model Selection

I implemented 3 different models with a 80% train 20% test split in hopes of finding the most accurate:



Null Model

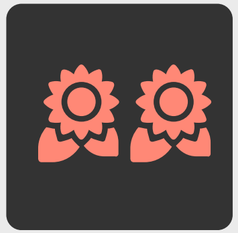
Predicting the majority class

PROS:

- Had 84.4% accuracy which was fairly high
- Easy to implement and quick to process

CONS:

- Only predicts the majority class so it is not the most accurate



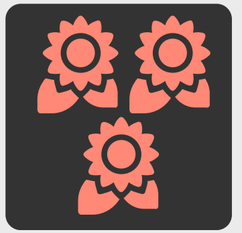
Simple Model

Decision Tree

PROS:

- Had 85.7% accuracy so was an impovement from null model
- Easy to implemenet and quick to process

CONS: Accuracy only improved from null modelby about 1%



Complex Model

Boosting

PROS:

- Had 89.6% accuracy (highest accuracy yet)
- Easy to implement and quick to process

CONS:

- Still was unable to crack the 90% accuracy threshold



Model Training

I began by creating a new train test set (70% train 30% test) to compare to the existing 80% train 20% test set

I then implemented the same model training on both data sets

80% train 20% test set



Decision Tree with Grid Search

- higher testing accuracy (89.99%)
- takes a very long time to run



Boosting with Randomized Search

- higher testing accuracy (89.67%)
- A bit challenging to implement (lots of errors to debug)

70% train 30% test set



Decision Tree with Grid Search

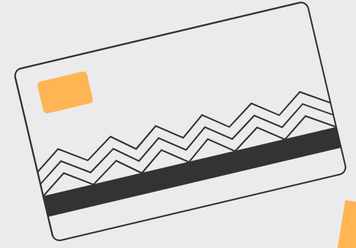
- higher testing accuracy (89.99%)
- takes a very long time to run



Boosting with Randomized Search

- higher testing accuracy (89.48%)
- A bit challenging to implement (lots of errors to debug)

Model Evaluation





- I evaluated all of the models for both of the data splits (80/20 & 70/30)
    - Each model recieved a respective classification report, confusion matrix, and AUC score

Classification Report highlights:

- The trained models for both splits had higher precision, recall, and F-1 scores compared to the basic models
- The boosting model with random search in the 70/30 split had the best scores overall, meaning it predicts the ”True” entries the best

Confusion Matrix Highlights:

- The complex model with random search was able to identify the true positives and the true negatives in the 70/30 train-test split the best, as the numbers in the top left corner, and the bottom right corner were higher than those for any other
- The null model classified the data the worst, as it didn't even identify any "True" positives (as it only predicts the majority class)

AUC Score Highlights:

- The simple model with grid search (trained decision tree) in the 70/30 train test split distinguishes between the "True" and "False" classes the best, as it has the highest score (0.78)
- The null model performs the worst as it has the lowest score(0.5), which kind of makes sense as it just predicts the majority class




Concluding Thoughts



- Both the simple model with grid search and the complex model with random search, both in the 70/30 train test split performed the best and would be the best models to implement

- If I had to pick between the two models, I would likely pick the complex model with random search as it seemed to be the best working model throughout the evaluation process regarding two of the three metrics (had the classification report and best organized confusion matrix)
  - The simple model had the best AUC score, which is why that model would also be considered

- I really wish one of the models had broken the 90% accuracy threshold
  - To make this happen, I would like to further implement data cleaning, as this data set was quite “messy” (lots of skewed features), and I think performing additional cleaning may improve the accuracy of all of the models I explored

- This has answered my question that yes, we can predict when a user’s web visit results in a transaction; however, it is much easier to answer the question when a user’s web visit does NOT result in a transaction

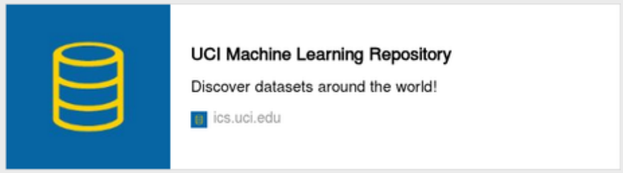
- This model could definitely be used by marketing teams to determine what attributes lead to transactions



Thank You







https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset


slidesgo